

THE UNIVERSITY OF ZAMBIA
SCHOOL OF HUMANITIES AND SOCIAL SCIENCES
2014/15 ACADEMIC YEAR FINAL EXAMINATIONS
DEM 1110: INTRODUCTION TO DEMOGRAPHY

TIME: THREE HOURS

INSTRUCTIONS: ANSWER ALL QUESTIONS IN SECTION A (COMPULSORY); ANY TWO QUESTIONS FROM SECTION B: AND ONE QUESTION FROM SECTION C (COMPULSORY)

SECTION A
ANSWER ALL QUESTIONS IN THIS SECTION.

1.
 - a) Which Muslim scholar is reputed to be the father of demography?
 - b) Who pioneered the use of statistical methods in evaluating demographic data?
 - c) Who constructed a primitive and crude form of a life table?
 - d) Which demographer predicted that the population was doubling every 25 years?
 - e) Which two famous historical figures rejected the Malthusian perspective?
 - f) In which year and at which conference did demography receive international recognition?
 - g) Which organization officially recognized demography as a discipline? Which year was this?
 - h) Who are the two famous figures who refined the logistic curve and applied it to human population growth?
 - i) At which university was the Office of Population Research established? Which year was this?
 - j) Which family founded the Population Council?
 - k) What was the disciplinary background of Raymond Pearl?
 - l) What was the disciplinary background of Alfred Lotka?
 - m) What was the disciplinary background of Frank Notestein?
 - n) What was the disciplinary background of Lowell Reed?
2. Which of the following best describes i) formal demography ii) population studies.
 - a) Execution and administration of censuses
 - b) Computation of life expectancy using life tables
 - c) Projection of the population
 - d) Estimation of mortality and fertility using incomplete data
 - e) A study of the relationship between changes in income and fertility and mortality
 - f) A study of the relationship between educational attainment and declining levels of fertility.
 - g) An analysis of fertility trends and changes in consumption patterns
 - h) A study of the aging of population and changes in voting patterns.
 - i) The analysis of changes in demographic variables mathematically.
3. On the basis of the scenarios below, briefly explain what would happen to:
 - a) If a country has high fertility and low mortality, the age structure will get older/younger resulting in a population that is adaptable/not adaptable to change and thus less innovative and receptive to technological change.
 - b) If a country has low fertility and low mortality, there will be rapid/slow population growth that will exhaust/match with the available resources thus exceeding/not exceeding its carrying capacity

4. Given the population in the table below, compute and interpret the following:
- The overall sex ratio
 - The dependency ratio
 - The percentage of the population below 20 years.
 - The percentage of the population above 15 years.
 - Briefly explain how the median age is used as a measure of the aging of the population and show whether Zambia has a young or an old population?
 - On the basis of the information in b), c) and d) is this population:
 - Young?
 - Intermediate?
 - Old?
 - In one or two sentences only, give one reason for your answer in e).

Age Group	Male	Female
0-1	198,388	201,122
1-4	396,775	402,243
5-9	553,193	564,638
10-14	511,845	516,401
15-19	454,345	485,050
20-24	329,925	380,836
25-29	248,186	284,597
30-34	210,486	218,863
35-39	146,636	150,014
40-44	126,510	139,051
45-49	104,291	109,868
50-54	92,170	99,813
55-59	74,276	66,513
60-64	59,054	56,498
65-69	40,674	33,224
70+	31,313	25,196
Total	3,578,067	3,733,927

5. Briefly explain, with examples where possible, what you understand by the following:
- Completed family size
 - Infant mortality rate
 - Arithmetic progression
 - Geometric progression
 - Doubling time of a country with 3.6 percent annual population growth rate.
 - De-jure method of enumeration
 - Inferential statistics
 - Retrospective approach
 - Passive method of registration system
 - Secondary data

SECTION B

ANSWER QUESTION 1 (COMPULSORY) AND ANY OTHER QUESTION BETWEEN 2 AND 3

1. Explain, with relevant examples, the role of demography in following situations:
 - a) Domestic investment planning.
 - b) Political planning.
 2. Marx and Engels believed that each economic system – capitalism and socialism – has its own laws of population. Briefly explain how these laws operated in:
 - a) The capitalist system
 - b) The socialist system
 3. Answer the following questions:
 - a. Give three (3) problems associated with the collection of social, economic and demographic data in Zambia?
 - b. Give three (3) inherent limitations of population registers in developing countries?
 - c. State three (3) reasons why it is important to evaluate socio-economic data before are used?
 - d. Explain the two (2) types of errors in data?
 - e. Briefly explain the Direct Method of detecting errors in social, economic and demographic data?
 - f. When evaluating demographic data using the Whipples index, what does an index of 100 mean?
 - g. What does the sharp fluctuations in the inter censal growth rates between sexes (male and female) in one census to another mean?
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SECTION C

ANSWER THIS QUESTION (COMPULSORY)

1. Use the table below to answer the following questions
 - a) What is the ratio of the urban residence households with a place for hand washing observed to that of rural residence with place for hand washing observed?
 - b) Comment on the percentage of households where place for washing hands was observed by residence (rural-urban)?
 - c) What is the proportion of the number of households in Muchinga province to the total number of households in Zambia?
 - d) Give two (2) important points to consider when displaying data on a graph or a chart?
 - e) What type of a graph would be appropriate to display percent distribution of the availability of cleansing agents (Soap and water, water and other cleansing

agents other than soap only, water only, soap but not water, cleansing agents other than soap only, no soap, no water no other cleansing agent, and missing)

- f) Comment on the results for the non-availability of water, no soap, no other cleansing agent by province in Zambia?

Table 2.6 Hand washing

Percentage of households in which the place most often used for washing hands was observed, and among households in which the place for hand washing was observed, percent distribution by availability of water, soap, and other cleansing agents, Zambia 2013-14

Background characteristic	Percentage of households where place for washing hands was observed	Number of households	Among households where place for hand washing was observed, percentage with:							Missing	Total	Number of households with place for hand washing observed
			Water and cleansing agent ² other than soap only	Water only	Soap but no water ³	Cleansing agent other than soap only ²	No water, no soap, no other cleansing agent					
Residence												
Urban	56.9	6,640	42.8	0.5	27.6	4.1	0.3	24.4	0.2	100.0	3,780	
Rural	28.2	9,280	17.3	2.5	24.6	5.4	3.0	46.2	0.9	100.0	2,615	
Province												
Central	23.1	1,472	29.8	0.6	39.9	0.9	0.5	27.2	1.1	100.0	339	
Copperbelt	65.9	2,455	41.7	1.8	22.8	4.6	0.9	28.1	0.2	100.0	1,619	
Eastern	30.1	1,938	20.9	0.9	44.7	1.4	0.3	30.3	1.4	100.0	583	
Lusaka	33.7	1,265	19.8	0.9	13.6	6.7	0.4	57.3	1.3	100.0	427	
Muchinga	60.3	2,925	39.1	0.6	30.3	3.7	0.0	26.2	0.2	100.0	1,765	
Northern	16.8	881	13.0	1.7	15.0	2.8	14.8	51.2	1.5	100.0	148	
North Western	29.2	1,269	21.2	1.7	9.8	13.1	7.0	46.5	0.8	100.0	370	
Southern	27.1	724	15.9	1.5	19.0	2.7	2.0	58.6	0.3	100.0	196	
Western	35.6	1,934	21.3	3.0	22.5	8.2	2.7	42.2	0.1	100.0	689	
Wealth quintile												
Lowest	23.0	3,514	10.3	1.4	23.0	4.4	3.0	56.6	1.2	100.0	808	
Second	25.7	3,055	13.5	2.0	22.4	6.3	3.8	50.7	1.2	100.0	787	
Middle	33.9	2,958	17.4	3.9	23.0	6.1	2.7	46.5	0.4	100.0	1,001	
Fourth	47.0	3,252	26.3	0.8	31.6	5.1	0.6	35.5	0.1	100.0	1,529	
Highest	72.3	3,141	57.5	0.2	27.0	3.2	0.0	11.7	0.3	100.0	2,270	
Total	40.2	15,920	32.4	1.3	26.4	4.6	1.4	33.3	0.5	100.0	6,396	

END OF EXAMINATION